Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Cotton Irrigation and Nutrient Management

1.1 Seasonal Water Requirements, Irrigation Scheduling, and Application Methods

- **Seasonal water requirements and crop demand:** Cotton (*Gossypium* spp.) requires 4 to 6 irrigations under conventional surface flooding and furrow systems, consuming 600 to 800 mm of water throughout its growth cycle depending on evaporative demand, soil texture, and seasonal rainfall distribution.
- **Pre-sowing irrigation (*Rauni*):** A heavy pre-sowing irrigation is mandatory before field preparation to thoroughly charge the deeper soil profile with moisture. This deep moisture reserve ensures rapid, uniform seed germination and robust early taproot penetration.
- **The "*Wattar*" condition for field operations:** Sowing and final seedbed preparation must take place precisely when the soil reaches the *Wattar* (workable moisture) condition after *Rauni*. At this moisture level, tractor machinery and implements can pass without compacting soil structure, while providing ideal inherent moisture for seed placement.
- **Post-sowing irrigation schedule:**
  - *First Irrigation Timing:* Delaying the first post-sowing irrigation until 6 to 7 weeks after sowing encourages deep vertical taproot extension into the subsoil. On light-textured, sandy soils or ridge-sown crops, the first irrigation may be advanced if early water stress is observed.
  - *Subsequent Irrigations:* Further water applications are scheduled at 2 to 3-week intervals based on crop growth requirements, soil texture, and weather parameters. Crop water stress must be evaluated during early morning hours to avoid false interpretations from midday temperature wilting.
  - *Critical Moisture Stages:* Water deficits must be strictly avoided during peak squaring, flowering, and boll development (July–August), as moisture stress triggers ethylene peaks and massive shedding of squares and young bolls.
  - *Pre-Harvest Water Cut-Off:* All irrigation activities must be terminated by the end of September (or 2–3 weeks before final harvest) to hasten natural boll drying, promote uniform boll cracking, and prevent delayed harvesting.
- **Irrigation application methods:** Water is applied through four primary irrigation methods:
  - *Flood Irrigation:* Conventional surface flooding across flat basins.

  - *Furrow Irrigation:* Water applied through alternate or every-furrow channels, saving up to 20–30% water compared to flood irrigation.
  - *Sprinkler Irrigation:* Overhead pressurized water delivery suitable for undulating or light sandy soils.
  - *Drip / Subsurface Drip Irrigation:* Precision micro-irrigation delivering water directly to the plant root zone, achieving higher water use efficiency, reducing weed growth between rows, and enabling uniform fertigation.

1.2 Waterlogging Sensitivity, Drainage Protocols, and Salinity Stress Mitigation

- **Waterlogging sensitivity and root suffocation:** Cotton is exceptionally sensitive to soil water stagnation, particularly during early seedling emergence and peak flowering. Water stagnation remaining for more than 24 hours causes root oxygen starvation (hypoxia/anoxia), restricts root respiration, induces root decay, halts nutrient absorption, and triggers sudden wilting or parawilt.
- **Field grading and drainage infrastructure:** Fields must be graded using Laser Land Levellers prior to sowing and provided with deep surface drainage ditches to evacuate excess rainwater rapidly after heavy rainstorms.
- **Post-drainage recovery protocols:**
  - Immediately after draining standing water, apply a foliar spray of 2% Potassium Nitrate (KNO_3 / 13:0:45) or 2% Urea solution at weekly intervals to restore plant vigor and nutrient uptake.
  - Perform light intercultivation or hoeing as soon as the soil surface dries to break the surface crust and restore rhizosphere aeration.
- **Soil salinity and poor water quality management:** High soil electrical conductivity (EC) or saline tube-well water induces osmotic stress, restricting root development and reducing yields. Recommended management precautions include:
  - Apply heavy pre-sowing *Rauni* using good-quality canal water to leach accumulated soluble salts below the root zone. Poor-quality tube-well water must never be used for pre-sowing irrigation.
  - Sow seeds on flat beds or inside the furrows of ridge-furrow systems rather than on ridge tops where salts accumulate via capillary rise.
  - Apply the first post-sowing irrigation using good-quality canal water.

1.3 Nutrient Uptake Dynamics and State-Wise Fertilizer Dosage Schedules

- **Nutrient uptake profile:** Cotton exhibits high seasonal demands for Nitrogen (N) and Potassium (K), exceeding 200 kg/ha NPK with peak uptake rates reaching up to 4 kg/ha/day during flowering and boll filling.
- **State Agricultural University (SAU) fertilizer recommendations (per acre basis):**

  - *Punjab:*
    * Hybrids of American and Desi cotton: 90 kg Urea, 27 kg DAP (or 75 kg SSP).
    * Bt varieties of American cotton: 80 kg Urea, 27 kg DAP.
    * Non-Bt varieties of American cotton: 65 kg Urea, 27 kg DAP.
    * Desi cotton varieties: 45 kg Urea.
    * Light soils: Apply 20 kg Muriate of Potash (MOP) per acre.
  - *Haryana:*
    * Hybrids of American and Desi cotton: 150 kg Urea, 50 kg DAP (or 150 kg SSP), 40 kg MOP.
    * Non-Bt varieties of American cotton: 75 kg Urea, 27 kg DAP (or 75 kg SSP).
    * Desi cotton varieties: 45 kg Urea.
  - *Rajasthan:*
    * Hybrids of American and Desi cotton: 130 kg Urea, 35 kg DAP (or 100 kg SSP), 14 kg MOP.
    * Non-Bt varieties of American cotton: 87 kg Urea, 35 kg DAP (or 100 kg SSP).
    * Desi cotton varieties: 78 kg Urea, 18 kg DAP (or 50 kg SSP).

- **Basal vs. Split Application Strategies:**
  - *Phosphorus and Potassium:* Drill all Phosphorus (DAP or SSP) and Potassium (MOP) as a basal application during final land preparation or at sowing.
  - *Nitrogen Splitting:* Apply 50% of recommended Nitrogen at thinning (25–30 DAS) and top-dress the remaining 50% at flower initiation. On low-fertility soils, apply the first 50% N dose at sowing instead of thinning.

- **Micronutrient supplementation:**
  - *Zinc Sulphate:* Apply 10 kg Zinc Sulphate heptahydrate (21%) or 6.5 kg Zinc Sulphate monohydrate (33%) per acre as a basal dose at sowing.
  - *Boron:* Apply 400 g Boron (4 kg borax) per acre at sowing in Boron-deficient soils.
  - *Magnesium Sulphate:* Apply 25 kg Magnesium Sulphate per hectare as a basal dose at sowing in deficient soils.

1.4 Fertilizer Application Techniques and Targeted Foliar Feeding

- **Fertilizer application methods in cotton fields:**
  - *Basal Broadcasting:* Surface application using tractor-operated broadcasters prior to final land preparation.
  - *Root-Zone Placement:* Drilling basal fertilizers near the root zone using tractor-operated intercultivator-cum-fertilizer drills.

  - *Top-Dressing:* Manual top-dressing (spot placement or row broadcasting) along plant rows after weeding operations.
  - *Foliar Feeding:* High-clearance tractor sprayers or knapsack sprayers applying dissolved water-soluble fertilizer solutions directly to the leaves.

- **Targeted foliar feeding schedules:** To prevent nutrient deficiencies during peak reproductive demand (July–August) and manage leaf reddening in Bt cotton:
  - *Potassium Nitrate (KNO_3 / 13:0:45):* Apply 3 to 5 foliar sprays of 2% solution at weekly intervals starting at flower initiation to enhance boll retention, boll weight, and drought resistance.
  - *Magnesium Sulphate ($MgSO_4$):* Apply 2 sprays of 1% solution (2.5 kg $MgSO_4$ in 250 L water per hectare) at 15-day intervals during full bloom and boll development.
  - *NPK (19:19:19) Foliar Spray:* Apply 1–2% 19:19:19 solution during boll development to support late-season canopy health.
  - *Urea / DAP Sprays:* Apply 2% Urea or 2% DAP foliar sprays following waterlogging or heavy rain to restore nitrogen levels.

- **Fertilizer safety guidelines:** Dissolve specified fertilizer quantities thoroughly in recommended water volumes before spraying. Avoid broadcasting Urea on wet foliage, and never re-spray leftover fluid onto previously treated foliage to prevent scorching.

1.5 Exhaustive Nutrient Deficiency Diagnostics and Corrective Measures

- **Nitrogen (N) Deficiency:**
  - *Symptoms:* Appears first on older lower leaves as uniform pale-green chlorosis, turning yellow and developing brown necrotic spots before advancing to upper foliage; plants become stunted with reduced sympodial branching.
  - *Correction:* Top-dress Urea @ 25–30 kg/acre followed by irrigation, or spray 2% Urea solution.

- **Phosphorus (P) Deficiency:**
  - *Symptoms:* Appears on lower mature leaves as dull, dark bluish-green foliage with reddish-purple pigmentation along leaf margins and interveinal areas, followed by premature leaf chlorosis, necrosis, and abscission.
  - *Correction:* Apply DAP or SSP as a basal drill application based on soil test values.

- **Potassium (K) Deficiency (Cotton Rust):**
  - *Symptoms:* Appears on lower mature leaves or upper canopy foliage during heavy boll loading. Leaves exhibit yellowish-white interveinal chlorosis, turning bronze or reddish-brown with curling margins and necrotic breakdown.

  - *Correction:* Soil application of MOP @ 20 kg/acre, or 3–5 foliar sprays of 2% Potassium Nitrate (KNO_3).

- **Magnesium (Mg) Deficiency:**
  - *Symptoms:* Lower mature leaves develop purplish-red or reddish interveinal coloration while leaf veins remain distinctly green.
  - *Correction:* Foliar spray of 1% Magnesium Sulphate ($MgSO_4$) twice at 15-day intervals during full bloom.

- **Zinc (Zn) Deficiency:**
  - *Symptoms:* Causes plant stunting, short internodes, bronzing of first true leaves, and "little leaf" with inward curling of top leaves.
  - *Correction:* Basal soil application of Zinc Sulphate heptahydrate (21%) @ 10 kg/acre at sowing.

- **Boron (B) Deficiency:**
  - *Symptoms:* Terminal growth dieback, crinkling of young leaves, and excessive drying and shedding of squares and young bolls.
  - *Correction:* Soil application of 400 g Boron (4 kg borax) per acre at sowing.

- **Calcium (Ca), Sulfur (S), and Iron (Fe) Deficiencies:**
  - *Calcium:* Crinkled mature leaves, yellowish-green upper foliage with purplish tinges, and drying of squares and bolls.
  - *Sulfur:* Light green to yellow chlorosis on younger top leaves while older leaves remain green.
  - *Iron:* Interveinal chlorosis and yellowish spots on young top leaves, turning pale yellow while older leaves remain normal.

1.6 Physiological Disorders Induced by Moisture and Nutritional Imbalance

- **Parawilt / Sudden Wilt Complex:**
  - *Causative Factors:* Occurs when heavy rain or over-irrigation saturates dry soil, followed by intense, hot sunshine. High-yielding Bt hybrids with heavy boll loads undergo a severe breakdown of the Soil-Plant-Atmosphere Continuum (SPAC), where transpirational demand exceeds root water uptake capacity. Aggravated by subsurface hardpans and poor root branching.
  - *Diagnostic Symptoms:* Rapid wilting, leaf drooping, chlorosis, bronzing/reddening, and premature shedding of squares and young bolls.
  - *Management Protocol:* Deep ploughing with a Mouldboard (MB) plough or sub-soiler at 1.0 m spacing before sowing; avoiding early over-irrigation; spraying 2% KNO_3; applying Cobalt Chloride @ 10 ppm or Sodium Benzoate @ 50 ppm, or root drenching with Copper Oxychloride (25 g) + Urea (200 g) per 10 L water within 24–48 hours of symptom appearance.

- **Leaf Reddening in Bt Cotton:**
  - *Causative Factors:* Triggered by Nitrogen and Magnesium deficiencies combined with low night temperatures (<15°C) during peak boll development, causing chlorophyll breakdown and red anthocyanin pigment accumulation.
  - *Management Protocol:* Apply Magnesium Sulphate ($MgSO_4$) @ 20–25 kg/ha to soil or spray 1% $MgSO_4$ + 2% KNO_3 + 1% Urea twice at 15-day intervals during full bloom.
- **Poor Taproot Development / Right-Angle Root Bending:**
  - *Causative Factors:* Dense subsurface soil hardpans cause taproots to bend horizontally at right angles, restricting vertical water and nutrient extraction.
  - *Management Protocol:* Cross-subsoiling with a chisel plough at 1.0 m spacing during land preparation.

1.7 Integrated Soil Health, Biofertilizers, and Seasonal Management Workflow

- **Pre-sowing soil health conditioning:** Conduct deep summer ploughing during May–June to expose soil to solar heat. Incorporate well-decomposed Farmyard Manure (FYM) @ 5 to 10 t/ha combined with Neem Cake @ 250 kg/ha during field preparation.
- **Microbial biofertilizers:** Treat seeds or soil with *Azotobacter*, Phosphate Solubilizing Bacteria (PSB), and *Trichoderma viride / T. harzianum* (@ 5–10 g/kg seed, or 10 kg/ha in 200 kg FYM) to enhance root health and nutrient solubilization.
- **Step-by-step seasonal management workflow:**
  1. *Pre-Sowing Phase:* Perform deep MB ploughing or subsoiling; test soil for NPK and micronutrients; apply FYM and Neem Cake; give heavy pre-sowing *Rauni* irrigation with canal water.
  2. *Sowing to Vegetative Phase (0–45 DAS):* Sow in *Wattar* soil condition; drill basal DAP/SSP, MOP, and Zinc Sulphate; delay first post-sowing irrigation to 6–7 weeks to encourage deep taproot extension; apply first Nitrogen split at thinning (25–30 DAS).
  3. *Reproductive Phase (45–120 DAS):* Top-dress remaining 50% N at flower initiation; maintain regular 2 to 3-week irrigation intervals without moisture stress; apply 3–5 foliar sprays of 2% KNO_3 and 2 sprays of 1% $MgSO_4$; drain standing water immediately after heavy rainfall.
  4. *Maturity Phase (120–180 DAS):* Terminate all irrigation by late September to encourage natural boll cracking and dry harvest.